

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	29 September 2026
Team ID	SWTID-2026-9956
Project Name	Auto Ticket Classification Using Flow Designer
Maximum Marks	4 Marks

Technical Architecture

The proposed solution is implemented on the ServiceNow platform to automate IT ticket creation, classification, dependent Category–Subcategory selection, assignment, and email notification. The architecture uses ServiceNow tables, fields, business rules, client-side configurations, server-side scripting, and notification mechanisms.

Table-1: Components & Technologies

S. No	Component	Description	Technology
1	User Interface	Provides the interface for users to create and view IT support tickets, enter issue descriptions, select ticket information, and track ticket status.	ServiceNow Service Portal / Platform UI, HTML, CSS, JavaScript
2	Application Logic-1	Automatically classifies IT tickets based on the issue description and determines the appropriate Category and Subcategory.	ServiceNow Business Rules, JavaScript
3	Application Logic-2	Implements dependent choice logic so that Subcategory options are dynamically displayed based on the selected Category.	ServiceNow Dictionary Configuration, Dependent Choices
4	Application Logic-3	Automatically sends an email notification to the Caller when a new IT ticket is created.	ServiceNow Notifications, Business Rules, Email Engine
5	Database	Stores structured IT ticket information including Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, and Assigned To.	ServiceNow Platform Database, Tables and Fields

6	Cloud Database	Provides centralized cloud-based storage for ticket records and configuration data.	ServiceNow Cloud Platform Database
7	File Storage	Supports storage of ticket-related attachments and supporting files when required.	ServiceNow Attachment Storage
8	External API-1	External integrations can be added in the future for additional IT service or ticket classification requirements.	ServiceNow REST API
9	External API-2	Supports future integration with external enterprise systems for exchanging ticket or user information.	REST / SOAP APIs
10	Machine Learning Model	The system can support automated classification of tickets based on issue descriptions. For the current implementation, classification can be handled using configured ServiceNow business logic and rules.	ServiceNow Business Rules / JavaScript; ML integration can be added in future
11	Infrastructure (Server / Cloud)	Hosts the application, database, business logic, configurations, and notification services.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1	Open-Source Frameworks	The solution primarily uses the ServiceNow platform and its built-in development capabilities. JavaScript is used for implementing application logic and automation.	JavaScript, ServiceNow Scripting APIs
2	Security	Access to ticket information is controlled using ServiceNow users, roles, groups, permissions, and access controls. Ticket data can be restricted according to organizational requirements.	ServiceNow ACLs, Roles, Authentication, Authorization

3	Scalable Architecture	The solution is designed using modular ServiceNow components such as tables, fields, business rules, notifications, and configurations. New categories, subcategories, users, groups, and automation rules can be added without redesigning the complete system.	ServiceNow Platform, Tables, Business Rules, Configuration
4	Availability	The application is deployed on the ServiceNow cloud platform, allowing authorized users to access the ticketing system through the ServiceNow interface.	ServiceNow Cloud Infrastructure
5	Performance	Server-side business logic is used for automation, while dependent choices reduce unnecessary user input and improve form usability. Automated notifications are generated through the ServiceNow notification mechanism.	ServiceNow Business Rules, Client Scripts, Dictionary Configuration, Notification Engine